

Wenjie Luo

Agent Application Engineer | Graduating Jan 2027

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PROFILE

Build Python and TypeScript agent applications with deterministic tools and human approval. Turn business workflows into APIs, state transitions, execution traces and automated acceptance tests.

EDUCATION

National University of Singapore | MSc Civil Engineering (Transportation) Aug 2025 - Jan 2027

Transport focus; research in transportation information and control.

Chang'an University | BEng Transportation Engineering Sep 2021 - Jun 2025

GPA 3.16/5.0; approximately top 15% of the cohort.

PROJECTS

Civil Buddy | Agent Runtime and Tools Jul 2026 - Present

- Built session scheduling, a tool engine, policy checks and recovery for 66 role workflows; required human confirmation for high-risk writes and retained failure audit records.
- Linked packing, confirmation and visualization; passed 128 deterministic evaluations. Separate shadow tests compare predefined steps with tool calling, including a keyless policy fallback.

Glass Box | Agent Observability Middleware Aug 23, 2026 - Present

- Built observability on the official starter: parent-child span waterfalls show tools, tokens, estimated costs and retries, with trace queries and OTLP/JSON export.
- Built keyless replay demos and policy tests for 22 bypass attempts and 11 allowed cases; Node 22/24 CI runs 150 tests in 21 files, type checks, builds and smoke tests.

EDA Copilot | Traceable Engineering Q&A Sep 2026

- Ran OpenROAD's nangate45/gcd example from RTL to GDSII; indexed reports, logs and workflow documentation into 1,215 chunks with source line references.
- Built BM25, TF-IDF and hybrid retrieval with cited excerpts; added CI checks for metric definitions, run identity and missing evidence.
- On self-authored offline tests, BM25 and hybrid both reached Hit@5 of 22/23; guardrails abstained correctly on all 12 refusal cases, versus none when disabled.

EXPERIENCE

Far East Facade | Project Engineer, Engineering Department Jun 2026 - Present

- Built a Python workflow to extract delivery batches and test results from supplier emails, supporting delivery plans for 15 material types.

Caixun Technology | Product Intern Jun 2025 - Aug 2025

- Moved the hard-to-find prize-draw entry from the page bottom to a side pop-up to improve visibility; defined requirements, designed interaction prototypes and followed development and testing.
- Independently wrote 3 PRDs and prototyped 5 core pages in Axure; proposals were accepted and scheduled for development. Tracked fixes and retesting, helping resolve over 10 issues.

SKILLS

Development: Python, TypeScript, React, Fastify/FastAPI, SQL, Git, CI

Agents: tool calling, session scheduling, human approval, tracing/OTLP, policy checks, recovery